A Note on an Old Proof of 'Quantum Toom's Rule'

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Abstract

This note examines the correctness of a proof of the 'Quantum Toom's rule' given in the old thesis [Ahn04], written almost twenty-five years ago. Although the presented ideas seem to point in the right direction, we found gaps that undermine the solidity of the work. Here, we present three of them. In a separate accompanying paper, we provide a candidate for a correct proof of the main theorem.

1 Introduction

The quantum Toom's rule has been mentioned extensively in the literature [Den+02], yet the question of the existence of a rigorously correct proof is not a consensus. One promising candidate is the work done by [Ahn04] in her Ph.D. thesis. We emphasize that her work is meaningful and contains clever concepts regardless of whether it's correct as a whole. Yet, as recent works have been relying on her result, it becomes obligatory to understand if the 'Quantum Toom's rule', as stated by her, is indeed valid. Here, we present three counterexamples to three arguments raised in her proof. The goal of the note is to be short and aimed, and therefore the text assumes that the reader is fluent in the proof details.

2 Failing of the Disobedient → Obedient Construction

3 Dense-Collection Probability Bound Relies on an Unjustified Completion Argument

4 Toom's Rule in Heigh Dimensions Doesn't Shrink Droplets

References

[Den+02] Eric Dennis et al. "Topological quantum memory". In: *Journal of Mathematical Physics* 43.9 (Sept. 2002), pp. 4452–4505. DOI: 10.1063/1.1499754. URL: https://doi.org/10.1063%2F1.1499754.

[Ahn04] Charlene Sonja Ahn. "Extending Quantum Error Correction: New Continuous Measurement Protocols and Improved Fault-Tolerant Overhead". Ph.D. Dissertation. California Institute of Technology, 2004. DOI: 10.7907/THD5-A335. URL: https://thesis.caltech.edu/1873/.